

Project Impact Statement: Automated Consumer Complaint Classification

Applications & Intended Benefits

This project develops an automated NLP system using DeBERTa-v3 to classify unstructured e-commerce and consumer service complaints across ten critical operational categories (e.g., delivery_shipping, billing, and fraud_unauthorized) within regional digital markets. By routing complex or high-risk complaints to consumer protection agencies and internal support teams automatically, the model significantly reduces resolution latency for vulnerable consumers.

Crucially, incorporating inverse class weighting directly addresses structural underrepresentation: severe but rare events such as unauthorized financial fraud are elevated immediately rather than buried under high-volume administrative inquiries. This improves consumer trust, mitigates financial exploitation in rapidly growing e-commerce ecosystems, and empowers regulatory ombudsmen with actionable data intelligence.

Societal Implications & Ethical Risks

While automated classification streamlines operations, deployment poses systemic risks to algorithmic fairness and consumer dignity if unmonitored.

- **Distributional Bias & Language Equity:** Because intermediate domain pre-training relies on scraped web reviews, the model risks inheriting localized slang, dialectal nuances, or demographic phrasing biases. False negatives on complaint severity for non-standard English phrasing could systematically deny redress to marginalized or non-native speaking communities.
- **Automation Bias & Autonomy:** Over-reliance on model outputs by support staff could lead to “automation bias,” where human agents blindly accept automated classifications without validating context, reducing consumer agency to appeal incorrect decisions.
- **Misclassification Escalation:** Mislabeling an urgent fraud_unauthorized claim as a general billing query defers financial intervention, directly risking unrecoverable monetary loss for low-income users.

Mitigation Initiatives & Governance Controls

To maximize societal utility and minimize harm, future deployments should implement a human-in-the-loop oversight framework:

1. **Uncertainty Thresholding:** Predictions where class probability falls below confidence thresholds (e.g., $P < 0.70$) or involve financial harm must automatically route to human agents for manual review rather than automated processing.
2. **Continuous Fairness Audits:** Periodic evaluations using disaggregated metrics across demographic proxies and text-length distributions must be conducted to prevent performance degradation on localized dialects.
3. **Transparent Redress Mechanisms:** Consumers must be explicitly notified when automated triage is used and provided a frictionless path to request human re-evaluation of their complaint classification.